Data Cleaning Practices and Their Impact on Univariate and Bivariate Analysis

Data cleaning is a crucial step in preparing datasets for accurate analysis, particularly as datasets grow in size and complexity. Errors such as misspellings, inconsistent formats, hidden characters, and incorrect data types can significantly distort analytical results if left unaddressed. Various tools like Excel, text editors, and OpenRefine help identify and correct these issues by standardizing entries, filling in missing values, and restructuring datasets into analyzable formats (Sunne, 2022; Groves, 2016).

For **univariate analysis**, which focuses on examining a single variable, proper data cleaning ensures that summary statistics (e.g., mean, frequency, distribution) are valid. Issues such as duplicate entries, incorrect numerical formatting (e.g., numbers stored as text), and outliers must be resolved beforehand. Without these corrections, measures of central tendency and dispersion may be misleading. Simple tools like Excel are often sufficient for these tasks when combined with systematic cleaning procedures (Elliott et al., 2006).

In contrast, **bivariate analysis** involves exploring relationships between two variables, such as correlation or regression. Data cleaning becomes even more critical here because inconsistencies across variables—such as mismatched categories, missing paired values, or inconsistent coding—can bias results or weaken statistical relationships. Tools like OpenRefine are especially useful for harmonizing categorical data and ensuring consistency across variables before conducting analysis (Groves, 2016).

Moreover, while Excel is widely used due to its accessibility, it may introduce errors when handling more complex analyses, especially if data cleaning is done manually. Studies suggest that spreadsheet-based analysis is prone to unnoticed mistakes and user overconfidence, which can affect both univariate and bivariate results. In contrast, programming environments like R provide more reliable and reproducible workflows, particularly for large datasets and advanced statistical procedures (Magar, 2025).

Generally, effective data cleaning improves the accuracy, reliability, and interpretability of both univariate and bivariate analyses, reducing errors and enhancing the overall quality of research findings.

References:

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